MODULE *Overwatch*

EXTENDS *Naturals*, *FiniteSets*

CONSTANTS *Names*, *MaxEdits*, *MaxAhead*, *RedBudget*, *Timeout*

ASSUME "timeout" $\notin$ *Names*

Modes $\triangleq$ {"advisory", "default", "strict"}
Versions $\triangleq$ 0 .. *MaxEdits*
Results $\triangleq$ SUBSET *Names*
TimedOut $\triangleq$ "timeout"
Outcomes $\triangleq$ SUBSET (*Names* $\cup$ {*TimedOut*})
NoMessage $\triangleq$ [*held* $\mapsto$ FALSE, *waking* $\mapsto$ FALSE, *failures* $\mapsto$ {}]
NotTold $\triangleq$ [*kind* $\mapsto$ "none", *fresh* $\mapsto$ {}]

VARIABLES
mode, *truth*, *head*,
busy, *from*, *quiet*, *installed*,
report, *baseline*, *wasFailing*,
outbox, *told*, *flush*,
turn, *toldThisTurn*, *promptQueued*,
knows, *done*,
elapsed, *redEdits*, *escalated*

vars $\triangleq$ ⟨*mode*, *truth*, *head*, *busy*, *from*, *quiet*, *installed*, *report*, *baseline*, *wasFailing*,
outbox, *told*, *flush*, *turn*, *toldThisTurn*, *promptQueued*, *knows*, *done*,
elapsed, *redEdits*, *escalated*⟩

limits $\triangleq$ ⟨*elapsed*, *redEdits*, *escalated*⟩

TypeOK $\triangleq$
$\land$ *mode* $\in$ *Modes*
$\land$ *truth* $\in$ [*Versions* $\to$ *Results*]
$\land$ *head* $\in$ *Versions*
$\land$ *busy* $\in$ BOOLEAN $\land$ *from* $\in$ *Versions* $\land$ *quiet* $\in$ BOOLEAN $\land$ *installed* $\in$ BOOLEAN
$\land$ *report* $\in$ [*known* : BOOLEAN , *at* : *Versions*, *failures* : *Outcomes*]
$\land$ *baseline* $\in$ [*known* : BOOLEAN , *failures* : *Outcomes*]
$\land$ *wasFailing* $\in$ BOOLEAN
$\land$ *outbox* $\in$ [*held* : BOOLEAN , *waking* : BOOLEAN , *failures* : *Outcomes*]
$\land$ *told* $\in$ [*kind* : {"none", "failed", "recovered"}, *fresh* : *Outcomes*]
$\land$ *flush* $\in$ BOOLEAN
$\land$ *turn* $\in$ {"idle", "running"}
$\land$ *toldThisTurn* $\in$ BOOLEAN $\land$ *promptQueued* $\in$ BOOLEAN
$\land$ *knows* $\in$ *Outcomes*
$\land$ *done* $\in$ [*known* : BOOLEAN , *at* : *Versions*, *mode* : *Modes*]
$\land$ *elapsed* $\in$ 0 .. *Timeout* $\land$ *redEdits* $\in$ 0 .. *RedBudget* $\land$ *escalated* $\in$ BOOLEAN

$Failing \triangleq report.known \land report.failures \neq \{\}$
$Regressed(f) \triangleq TimedOut \in f \lor (baseline.known \land \neg(f \subseteq baseline.failures))$
$Fresh \triangleq report.known \land report.at = head \land \neg quiet \land \neg busy$

$Guard(m) \triangleq$
  CASE $m =$ "advisory" $\rightarrow$ FALSE
    $\square$ $m =$ "default" $\rightarrow Failing \land Regressed(report.failures)$
    $\square$ $m =$ "strict" $\rightarrow Failing \lor \neg Fresh$

$Ahead \triangleq$ IF $report.known$ THEN $head - report.at$ ELSE $head$

$CanEdit(m) \triangleq$
  CASE $m =$ "advisory" $\rightarrow$ TRUE
    $\square$ $m =$ "default" $\rightarrow Ahead < MaxAhead$
    $\square$ $m =$ "strict" $\rightarrow Ahead < MaxAhead \land redEdits < RedBudget$

$Init \triangleq$
  $\land\ mode \in Modes$
  $\land\ truth \in [Versions \rightarrow Results]$
  $\land\ head = 0$
  $\land\ busy =$ TRUE $\land\ from = 0 \land\ quiet =$ FALSE $\land\ installed =$ FALSE
  $\land\ report = [known \mapsto$ FALSE, $at \mapsto 0$, $failures \mapsto \{\}]$
  $\land\ baseline = [known \mapsto$ FALSE, $failures \mapsto \{\}]$
  $\land\ wasFailing =$ FALSE
  $\land\ outbox = NoMessage \land told = NotTold \land flush =$ FALSE
  $\land\ turn =$ "idle" $\land\ toldThisTurn =$ FALSE $\land\ promptQueued =$ FALSE
  $\land\ knows = \{\}$
  $\land\ done = [known \mapsto$ FALSE, $at \mapsto 0$, $mode \mapsto$ "default"$]$
  $\land\ elapsed = 0 \land redEdits = 0 \land escalated =$ FALSE

$Queue(signature, message) \triangleq$
  IF $told = signature \land \neg outbox.held$
  THEN UNCHANGED $\langle outbox, told, flush\rangle$
  ELSE $\land\ told' = signature$
       $\land\ outbox' = message$
       $\land\ flush' =$ TRUE

$Deliver(v, f) \triangleq$
  LET $b \triangleq$ IF $baseline.known$ THEN $baseline.failures$ ELSE IF $head = 0$ THEN f ELSE $\{\}$
      $isRegression \triangleq \neg(f \subseteq b)$
  IN $\land\ report' = [known \mapsto$ TRUE, $at \mapsto v$, $failures \mapsto f]$
     $\land\ baseline' = [known \mapsto$ TRUE, $failures \mapsto b]$
     $\land\ wasFailing' = isRegression$
     $\land\ redEdits' =$ IF $isRegression$ THEN $redEdits$ ELSE 0
     $\land$ IF $isRegression$
         THEN $Queue([kind \mapsto$ "failed", $fresh \mapsto f \setminus b], [held \mapsto$ TRUE, $waking \mapsto$ TRUE, $failures \mapsto f])$
       ELSE IF $\neg wasFailing$

THEN UNCHANGED $\langle outbox, told, flush\rangle$
ELSE IF $outbox.held$
THEN $\land\ outbox' = NoMessage$
$\land\ told' = NotTold$
$\land$ UNCHANGED $flush$
ELSE $Queue([kind \mapsto$ "recovered", $fresh \mapsto \{\}], [held \mapsto$ TRUE, $waking \mapsto$ FALSE, $failures \mapsto f])$

$Edit \triangleq$
$\land\ turn =$ "running"
$\land\ head < MaxEdits$
$\land\ CanEdit(mode)$
$\land\ head' = head + 1$
$\land\ quiet' =$ TRUE
$\land\ redEdits' =$ IF $wasFailing \land mode =$ "strict" THEN $redEdits + 1$ ELSE $redEdits$
$\land$ UNCHANGED $\langle mode, truth, busy, from, installed, report, baseline, wasFailing,$
$outbox, told, flush, turn, toldThisTurn, promptQueued, knows, done,$
$elapsed, escalated\rangle$

$QuietElapses \triangleq$
$\land\ quiet$
$\land\ installed$
$\land\ quiet' =$ FALSE
$\land\ busy' =$ TRUE
$\land\ from' = head$
$\land\ elapsed' = 0$
$\land$ UNCHANGED $\langle mode, truth, head, installed, report, baseline, wasFailing,$
$outbox, told, flush, turn, toldThisTurn, promptQueued, knows, done,$
$redEdits, escalated\rangle$

$Tick \triangleq$
$\land\ busy$
$\land\ elapsed < Timeout$
$\land\ elapsed' = elapsed + 1$
$\land$ UNCHANGED $\langle mode, truth, head, busy, from, quiet, installed, report, baseline, wasFailing,$
$outbox, told, flush, turn, toldThisTurn, promptQueued, knows, done,$
$redEdits, escalated\rangle$

$RunEnds \triangleq$
$\land\ busy$
$\land\ \exists\, v \in from\ ..\ head : Deliver(v, truth[v])$
$\land\ busy' =$ FALSE
$\land\ installed' =$ TRUE
$\land$ UNCHANGED $\langle mode, truth, head, from, quiet, turn, toldThisTurn, promptQueued, knows, done,$
$elapsed, escalated\rangle$

$TimesOut \triangleq$

$\land$ *busy*
$\land$ *elapsed* $=$ *Timeout*
$\land$ *report*$'$ $=$ [*known* $\mapsto$ TRUE, *at* $\mapsto$ *from*, *failures* $\mapsto$ $\{$*TimedOut*$\}$]
$\land$ *wasFailing*$'$ $=$ TRUE
$\land$ *Queue*([*kind* $\mapsto$ "failed", *fresh* $\mapsto$ $\{$*TimedOut*$\}$], [*held* $\mapsto$ TRUE, *waking* $\mapsto$ TRUE, *failures* $\mapsto$ $\{$*TimedO*
$\land$ *busy*$'$ $=$ FALSE
$\land$ *installed*$'$ $=$ TRUE
$\land$ UNCHANGED $\langle$*mode*, *truth*, *head*, *from*, *quiet*, *baseline*, *turn*, *toldThisTurn*, *promptQueued*, *knows*,
elapsed, *redEdits*, *escalated*$\rangle$

Held $\triangleq$
IF *turn* $=$ "running" THEN *toldThisTurn* ELSE *promptQueued* $\lor$ *escalated* $\lor$ $\neg$*outbox.waking*

Flush $\triangleq$
$\land$ *flush*
$\land$ *flush*$'$ $=$ FALSE
$\land$ IF *outbox.held* $\land$ $\neg$*Held*
THEN $\land$ *outbox*$'$ $=$ *NoMessage*
$\land$ *knows*$'$ $=$ *outbox.failures*
$\land$ IF *turn* $=$ "running"
THEN *toldThisTurn*$'$ $=$ TRUE $\land$ UNCHANGED *promptQueued*
ELSE *promptQueued*$'$ $=$ TRUE $\land$ UNCHANGED *toldThisTurn*
ELSE UNCHANGED $\langle$*outbox*, *knows*, *toldThisTurn*, *promptQueued*$\rangle$
$\land$ UNCHANGED $\langle$*mode*, *truth*, *head*, *busy*, *from*, *quiet*, *installed*, *report*, *baseline*, *wasFailing*,
told, *turn*, *done*$\rangle$
$\land$ UNCHANGED *limits*

TurnStarts $\triangleq$
$\land$ *turn* $=$ "idle"
$\land$ *turn*$'$ $=$ "running"
$\land$ *toldThisTurn*$'$ $=$ *promptQueued*
$\land$ *promptQueued*$'$ $=$ FALSE
$\land$ *flush*$'$ $=$ TRUE
$\land$ *escalated*$'$ $=$ FALSE
$\land$ *redEdits*$'$ $=$ IF *escalated* THEN 0 ELSE *redEdits*
$\land$ UNCHANGED $\langle$*mode*, *truth*, *head*, *busy*, *from*, *quiet*, *installed*, *report*, *baseline*, *wasFailing*,
outbox, *told*, *knows*, *done*, *elapsed*$\rangle$

WokenTurnStarts $\triangleq$ *promptQueued* $\land$ $\neg$*escalated* $\land$ *TurnStarts*

TurnCompletes $\triangleq$
$\land$ *turn* $=$ "running"
$\land$ IF *Guard*(*mode*)
THEN $\land$ *toldThisTurn*$'$ $=$ TRUE
$\land$ *knows*$'$ $=$ IF *Failing* THEN *report.failures* ELSE *knows*
$\land$ UNCHANGED $\langle$*turn*, *flush*, *done*$\rangle$

ELSE $\land\ turn' = \text{“idle”}$
$\land\ flush' = \text{TRUE}$
$\land\ done' = [known \mapsto \text{TRUE},\ at \mapsto head,\ mode \mapsto mode]$
$\land$ UNCHANGED $\langle toldThisTurn,\ knows\rangle$
$\land$ UNCHANGED $\langle mode,\ truth,\ head,\ busy,\ from,\ quiet,\ installed,\ report,\ baseline,\ wasFailing,$
$outbox,\ told,\ promptQueued\rangle$
$\land$ UNCHANGED $limits$

$Escalates \triangleq$
$\land\ mode = \text{“strict”}$
$\land\ turn\ = \text{“running”}$
$\land\ wasFailing$
$\land\ redEdits = RedBudget$
$\land\ turn' = \text{“idle”}$
$\land\ escalated' = \text{TRUE}$
$\land\ knows' = report.failures$
$\land$ UNCHANGED $\langle mode,\ truth,\ head,\ busy,\ from,\ quiet,\ installed,\ report,\ baseline,\ wasFailing,$
$outbox,\ told,\ flush,\ toldThisTurn,\ promptQueued,\ done,\ elapsed,\ redEdits\rangle$

$Next \triangleq Edit \lor QuietElapses \lor Tick \lor RunEnds \lor TimesOut \lor Flush \lor TurnStarts \lor TurnCompletes \lor Es$

$Spec \triangleq$
$\land\ Init \land \Box[Next]_{vars}$
$\land\ \mathrm{WF}_{vars}(QuietElapses) \land \mathrm{WF}_{vars}(Tick) \land \mathrm{WF}_{vars}(TimesOut) \land \mathrm{WF}_{vars}(Flush)$
$\land\ \mathrm{WF}_{vars}(WokenTurnStarts) \land \mathrm{WF}_{vars}(TurnCompletes) \land \mathrm{WF}_{vars}(Escalates)$

$ModesNest \triangleq$
$\land\ Guard(\text{“advisory”}) \Rightarrow Guard(\text{“default”})$
$\land\ Guard(\text{“default”}) \Rightarrow Guard(\text{“strict”})$
$\land\ CanEdit(\text{“strict”}) \Rightarrow CanEdit(\text{“default”})$
$\land\ CanEdit(\text{“default”}) \Rightarrow CanEdit(\text{“advisory”})$

$EditsStayCloseToResults \triangleq$
$mode \neq \text{“advisory”} \Rightarrow Ahead \leq MaxAhead$

$RedEditsAreBounded \triangleq$
$mode = \text{“strict”} \Rightarrow redEdits \leq RedBudget$

$RunsEnd \triangleq$
$busy \leadsto \neg busy$

$EditingResumes \triangleq$
$\neg CanEdit(mode) \leadsto (CanEdit(mode) \lor escalated)$

$StrictCompletesOnlyOnVerifiedGreen \triangleq$
$done.known \land done.mode = \text{“strict”} \Rightarrow truth[done.at] = \{\}$

$BaselineNeverAbsorbsEdits \triangleq$

$$baseline.known \Rightarrow baseline.failures \subseteq truth[0]$$

$$EveryEditIsChecked \triangleq$$
$$\Diamond\Box(report.known \land report.at = head)$$

$$RegressionsReachAgent \triangleq$$
$$\Diamond\Box((truth[head] \setminus truth[0]) \subseteq knows \lor TimedOut \in knows \lor escalated)$$